Homework 1: Foundations of Generative Modeling (Lectures 0–6)

CS25 – Large Foundation Models

Beijing Institute of Mathematical Sciences and Applications

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Beijing Institute of Mathematical Sciences and Applications

Homework at a Glance

- **Goal:** connect the mathematical objectives (Lectures 0–6) to working implementations and honest empirical comparisons.
- **You will submit:**
  - **Report (PDF, 3–6 pages)**: short derivations + experiments + discussion.
  - **Code (zip or repo link)**: training + sampling + plotting scripts/notebooks.
  - **Reproducibility file**: README with exact run commands + environment.
- **Recommended time:** 6–10 hours (depending on chosen track).

Learning Objectives (Mapped to Lectures 0–6)

After this homework, you should be able to:

- **Lecture 0:** summarize the "five families" view and trade-offs (train vs sample, likelihood vs implicit).
- **Lecture 1:** write down the diffusion forward process and explain score/denoising objectives.
- **Lecture 2:** explain flow matching as learning a velocity field and connecting it to an ODE transport.
- **Lecture 3:** derive the VAE ELBO and implement reparameterization.
- **Lecture 4:** apply change of variables and compute exact log-likelihood for a flow.
- **Lecture 5:** analyze GAN minimax, optimal discriminator, and practical instability.
- **Lecture 6:** factorize a joint distribution autoregressively and explain causal masking in Transformers.

Rules & Academic Integrity

- **Work policy:** you may discuss ideas, but your derivations, code, and report must be your own.
- **External code:** allowed only for *utility* (e.g., dataloaders, plotting). If you borrow model code, you must:
  - cite it clearly (URL + what you used),
  - explain it in your own words,
  - and still implement **at least one core component yourself** (e.g., objective, sampling loop, masking).
- **What we grade:** correctness + clarity + honest evaluation (not "best samples on earth").

Deliverables Checklist (1/2)

Submit these three items: `report.pdf`, `code/` (or notebooks), `README.md`.

Report PDF

- Part A answers (A0–A6 + 2 of A7–A10)
- chosen Track description (what you implemented)
- results: figures + tables + short discussion
- comparison table across families (Part C)

Code

- training + sampling entrypoints
- evaluation/visualization code
- fixed random seed + hyperparameters (config or args)
- clear folder structure (one command to reproduce key results)

Deliverables Checklist (2/2)

README.md must include:

- environment setup (`pip install -r requirements.txt` or equivalent)
- exact commands to reproduce:
  - one main training run
  - one sampling run that saves a grid/plot
  - one evaluation/plot that generates your key figure/table
- machine details (CPU/GPU) + runtime estimate (rough)

Repro rule: if we cannot reproduce your main figure/table, you lose most reproducibility points.

Part A: Theory (Make It Rigorous)

Part A is graded like a short math assignment. Your answers should look like clean lecture notes:

- **State assumptions** (data distribution, latent variable model, differentiability, invertibility, etc.).
- **Show intermediate steps** for each derivation (no "it is known that" jumps).
- **Name the object you optimize** (what expectation, over which random variables).
- **Explain meaning** of each term (1 sentence per term is enough).
- **Conclude with a takeaway** (what the derivation implies for training/sampling behavior).

Write-up format: answers A0–A10 with clear numbering (A7–A10 are the harder extensions; see slides).

Part A: What Counts as a Complete Answer

For each question, we expect:

- **(i) Math**: correct formula(s) with correct conditioning, distributions, and constants where needed.
- **(ii) A short proof/derivation**: 5–20 lines, depending on the question.
- **(iii) Interpretation**: 3–8 sentences connecting to the lecture (trade-offs, failure modes).

Minimum rigor rule: if you use a known identity (e.g., Jensen, change-of-variables, KL algebra), you must either (a) prove it briefly or (b) state it clearly and show how you apply it.

Part A: Hardness Upgrade (How We Will Grade)

- **Core understanding (A0–A6):** do you know the key objectives and why they work?
- **Deeper understanding (A7–A10):** can you connect objectives *across* families and reason about failure cases?
- **Common deductions:**
  - missing conditioning / wrong distributions,
  - stating final results without showing steps,
  - mixing up likelihood vs. implicit models,
  - confusing training objective with sampling procedure.

A0: The "Five Families" Comparison Table (Lecture 0)

Task. Fill in a 5-row comparison table for: **Autoregressive**, **VAE**, **GAN**, **Normalizing Flow**, **Diffusion** ***or*** **Flow Matching**.

Columns (required):

- training objective (1 line math),
- likelihood: exact / bound / none,
- sampling: one-pass / iterative / sequential,
- computational profile: train cost vs sample cost (qualitative),
- one typical failure mode (e.g., mode collapse, blurry samples, slow sampling, exposure bias, posterior collapse).

What to hand in: one table + 3–5 sentences summarizing *your* takeaway trade-off.

(a) Forward process. Write a standard Gaussian noising process, e.g.

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

or an equivalent Markov form $q(x_t \mid x_{t-1})$.

(b) Training signal. Choose one and write the loss precisely:

- score matching ($s_\theta(x_t, t) \approx \nabla_x \log p_t(x_t)$), or
- ϵ-prediction ($\epsilon_\theta(x_t, t) \approx \epsilon$), or
- x_0-prediction ($\hat{x}_{0,\theta}(x_t, t) \approx x_0$).

(c) Explanation (short). In 3–6 sentences, explain why gradients of $\log p(x)$ (scores) avoid normalization constants in energy-based forms.

A1 (Optional Challenge): Sampling and Cost

(d) Sampling loop. In 5–10 lines of pseudocode, write the high-level reverse sampling procedure (iterating $t = T \to 0$).

(e) Cost discussion. Explain (2–4 sentences) why diffusion often has:

- relatively stable training, but
- slower sampling (many steps),

and name one common acceleration idea (DDIM, fewer steps, distillation/consistency, ODE sampling, etc.).

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A2: Flow Matching as Learning a Velocity Field (Lecture 2)

(a) Core equation. State the ODE transport:

$$\frac{dx_t}{dt} = v_\theta(x_t, t), \qquad x_0 \sim p_0, \quad t \in [0,1].$$

(b) Continuity equation. Write the PDE relation (strong or weak form) linking v to marginals p_t.

(c) Regression view. Suppose you have training pairs $(x_t, \dot{x}_t)$ from a chosen coupling/path. Write the MSE loss and explain why the optimal predictor is a conditional expectation:

$$v^\star(x,t) = \mathbb{E}[\dot{x}_t \mid x_t = x, t].$$

What to hand in: (a)–(c) with 5–8 sentences total explanation.

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A3: Derive the ELBO (Lecture 3)

(a) Setup. Define $p_\theta(x, z) = p_\theta(x \mid z)p(z)$ and an approximate posterior $q_\phi(z \mid x)$.

(b) Derivation. Starting from $\log p_\theta(x)$, derive:

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \mathrm{KL}(q_\phi(z \mid x) \,\|\, p(z))\,.$$

(c) Interpretation. Label which term is reconstruction and which is regularization/prior matching.

(d) Reparameterization. Write the reparameterization for Gaussian latents and explain (2–3 sentences) why it helps gradients.

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A3 (Optional Challenge): Posterior Collapse

Explain **posterior collapse** (what it is, and when it happens) in 5–8 sentences.

Must include:

- what you would observe in the KL term,
- why a powerful decoder can cause it,
- one mitigation (KL annealing, free bits, weaker decoder, β-VAE, etc.).

A4: Change of Variables and Tractable Jacobians (Lecture 4)

(a) Formula. For an invertible map $x = T_\theta(z)$ with $z \sim p_Z$, write:

$$\log p_\theta(x) = \log p_Z(z) + \log \left| \det \frac{\partial T_\theta^{-1}(x)}{\partial x} \right| .$$

(b) Composition. For K layers, write the sum of log-determinants.

(c) Architecture reason. Explain what structure makes determinants easy:

- triangular Jacobians (autoregressive flows), or
- coupling layers (block triangular Jacobians),

and explicitly state what becomes "cheap" to compute.

A5: GAN Minimax and Optimal Discriminator (Lecture 5)

(a) Objective. Write the minimax GAN objective:

$$\min_G \max_D \; \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))].$$

(b) Optimal D. For fixed G, derive:

$$D_G^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}.$$

(c) Divergence link. Substitute D_G^* and show the Jensen–Shannon divergence form (you may skip constant terms if you state them).

What to hand in: derivation steps (not just the final line) + 3–6 sentences on what this says about convergence.

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A5 (Optional Challenge): Training Instability

Give **two** reasons GAN training is unstable in practice (e.g., non-convex game dynamics, discriminator too strong/weak, vanishing gradients, support mismatch).

Then: name **one** stabilization idea and briefly justify it:

- non-saturating loss, WGAN, gradient penalty, spectral norm, EMA, etc.

(a) Factorization. For a sequence $x_{1:T}$, write:

$$p(x_{1:T}) = \prod_{t=1}^{T} p(x_t \mid x_{<t}) \quad \Longleftrightarrow \quad \log p(x_{1:T}) = \sum_{t=1}^{T} \log p(x_t \mid x_{<t}).$$

(b) MLE loss. Write the negative log-likelihood (cross-entropy) training objective for a dataset.

(c) Sampling. Explain (3–5 sentences) why sampling is sequential and how this differs from one-pass flows.

A6: Causal Masking in Self-Attention (Lecture 6)

(a) Attention. Write the (scaled) dot-product attention:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V$$

(you may omit multi-head details).

(b) Mask. Define a causal mask M that prevents attending to future tokens (state what entries are $-\infty$ vs 0).

(c) Reasoning. In 4–8 sentences, explain how masking enforces dependence only on $x_{<t}$ and connects to the AR factorization in A6(a).

A7–A10: Harder Theory Extensions (Required)

These are the harder questions. Answer **at least 2 out of A7–A10** (you may answer all for bonus).

Why: these force you to connect the lectures, not just restate definitions.

What to hand in: for each chosen question, include (i) derivation/proof sketch, (ii) interpretation, (iii) one concrete implication for practice.

A7: Diffusion — Relating ϵ-Prediction to Score (Lecture 1)

Assume the standard Gaussian corruption

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

(a) Derive the conditional score $\nabla_{x_t} \log q(x_t \mid x_0)$ in closed form.

(b) Show the algebraic relationship between the score and the noise:

$$\nabla_{x_t} \log q(x_t \mid x_0) = -\frac{1}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon.$$

(c) Use (b) to explain (with equations) why training an $\epsilon_\theta(x_t, t)$ network can be viewed as a form of score learning (up to scaling).

(d) Interpretation: in 5–8 sentences, explain what this says about "why ϵ-prediction works" and what changes if the noise is not Gaussian.

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A8: Flow Matching — Continuity Eq. & Log-Density Dynamics (Lectures 2 & 4)

Let x_t follow the ODE $\dot{x}_t = v(x_t, t)$ and let p_t be the density of x_t. **(a)** Starting from the continuity equation

$$\partial_t p_t(x) + \nabla \cdot (p_t(x) v(x, t)) = 0,$$

derive the **log-density** evolution along trajectories:

$$\frac{d}{dt} \log p_t(x_t) = -\nabla \cdot v(x_t, t).$$

(b) Explain why (a) is the key identity behind continuous normalizing flows (CNFs).

(c) In 5–8 sentences, compare the training signal in flow matching (regress a velocity) vs. in diffusion (learn a score) and discuss one scenario where one signal is easier to obtain than the other.

A9: VAE — ELBO Gap and What KL Is Really Measuring (Lecture 3)

Let $p_\theta(x, z) = p_\theta(x \mid z)p(z)$ and $q_\phi(z \mid x)$. **(a)** Prove the "ELBO gap" identity:

$$\log p_\theta(x) - \mathrm{ELBO}(x) = \mathrm{KL}(q_\phi(z \mid x) \parallel p_\theta(z \mid x)) \geq 0.$$

(b) Expand the dataset-averaged KL term $\mathbb{E}_{p_{\text{data}}(x)}\mathrm{KL}(q_\phi(z \mid x) \| p(z))$ into a sum involving **mutual information** $I_q(x; z)$ and a marginal KL $\mathrm{KL}(q_\phi(z) \| p(z))$ (state your definition of $q_\phi(z)$).

(c) Interpretation: explain how (b) clarifies posterior collapse and why it is linked to the decoder strength.

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A10: GAN — Why Gradients Can Vanish (Lecture 5)

Consider the original GAN with optimal discriminator $D_G^*(x)$. **(a)** Assume p_{data} and p_g have (approximately) disjoint support early in training. Use $D_G^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x)+p_g(x)}$ to argue what values D_G^* takes on real vs fake regions.

(b) Using (a), explain (with a short derivative argument) why the generator gradient can become very small for the minimax loss.

(c) Propose one fix and justify it mathematically at a high level:

- non-saturating loss, or
- Wasserstein GAN idea (name the key theorem/constraint: 1-Lipschitz critic).

Part B: Choose One Track

Pick **exactly one** implementation track (B1/B2/B3). Each track asks you to implement and evaluate **at least two model families** from Lectures 1–6.

Track	**Summary**
B1: Toy 2D	Implement **Flow** (Lecture 4) *and* **GAN** (Lecture 5) on a 2D mixture (fast, great for debugging).
B2: MNIST	Implement **VAE** (Lecture 3) *and* one of: **Diffusion** (Lecture 1) or **GAN** (Lecture 5).
B3: Tiny text	Implement a **Transformer LM** (Lecture 6) *and* a baseline (e.g., bigram/MLP autoregressive) and compare.

Common Requirements for All Tracks

- **Reproducibility:** fix a random seed; log hyperparameters.
- **Training curve:** show at least one meaningful curve (loss, NLL, reconstruction, discriminator metrics).
- **Sampling:** generate and visualize at least 64 samples per model.
- **Comparison table:** include a table comparing the two chosen families on:
  - training time, sampling time, stability, and qualitative sample diversity.

Track B1: Dataset & Target

Dataset: 2D mixture of Gaussians (e.g., 8 Gaussians in a circle) or "two moons".

- Implement a **normalizing flow** (Lecture 4) that allows exact log-likelihood.
- Implement a **GAN** (Lecture 5) and compare coverage vs mode collapse.

Plots to include:

- scatter of real data vs generated samples,
- density heatmap for the flow (optional but recommended),
- training curves (NLL for flow; discriminator/generator losses for GAN).

Track B1: Minimal Model Specs (Suggested)

- **Flow:** 6–12 coupling layers; MLP conditioner; base $\mathcal{N}(0, I)$.
- **GAN:** MLP generator + MLP discriminator; use non-saturating loss; consider gradient penalty as a stability add-on (optional).
- **Evaluation:** for GAN, report a simple *coverage metric* (e.g., cluster assignment proportions) + show multiple random seeds.

Track B2: Dataset & Target

Dataset: MNIST (28×28 grayscale) via `torchvision.datasets.MNIST`.

- Implement a **VAE** (Lecture 3) with convolutional encoder/decoder (or MLP if you prefer).
- Implement **either**:
  - a small **diffusion model** (Lecture 1) with a tiny U-Net, *or*
  - a simple **DCGAN-style** model (Lecture 5).

Must include: sample grids + a short note on typical failure mode (blurry VAE vs instability or slow sampling).

Track B2: Minimal Model Specs (Suggested)

- **VAE:** latent dim 16–64; report reconstruction term and KL term separately.
- **Diffusion:** 50–200 steps (small); noise schedule of your choice; train $\epsilon_\theta(x_t, t)$.
- **GAN:** monitor mode collapse (visualize samples over time); try label smoothing or gradient penalty (optional).

Track B3: Dataset & Target

Dataset: a small text corpus (you choose; keep it under ∼5MB).

- Implement an **autoregressive Transformer** (Lecture 6) with causal masking.
- Implement a **baseline** autoregressive model (bigram, n-gram, or MLP with a fixed context window).

Must include:

- training curves (NLL/perplexity),
- samples at different temperatures,
- short explanation of why masking is needed (connect to factorization).

Track B3: Minimal Model Specs (Suggested)

- **Tokenizer:** character-level is fine (fast + simple).
- **Transformer:** 2–4 layers, 4 heads, d_model 128–256.
- **Evaluation:** report validation NLL/perplexity; show 3 sample generations with different seeds.

Part C: Comparison Table (Required)

Include a table comparing your two implemented families. Suggested columns:

- **Objective** (NLL / ELBO / minimax / denoising / regression).
- **Likelihood?** exact / bound / none.
- **Sampling** one pass / iterative ODE-SDE / autoregressive.
- **Stability issues** and what you observed.
- **Best use-case** based on your results and Lectures 0–6.

Part C: What We Look For in the Discussion

- **Honesty**: show failures; explain them using lecture concepts (mode collapse, posterior collapse, slow sampling, exposure bias, etc.).
- **Ablation (small)**: change one hyperparameter and report the effect (e.g., latent dim, number of flow layers, diffusion steps, learning rate).
- **Trade-offs**: connect your measurements to the Lecture 0 taxonomy.

Submission Format

- Submit a single **zip** (or a repo link) containing:
  - `report.pdf`
  - `code/` (or notebooks)
  - `README.md` with exact commands
  - `requirements.txt` or environment spec
- **Naming:** `CS25_HW1_<YourName>.zip`
- **Late policy:** fill in according to course policy.

Grading Rubric

Category	**What we grade**
Theory (35%)	correctness, clean derivations, correct connections to Lectures 0–6
Implementation (35%)	correct training + sampling, sensible code structure, reproducibility
Experiments (20%)	plots, samples, at least one ablation, honest comparison
Communication (10%)	report clarity, tables/figures labeled, limitations stated

FAQ

- **Do I need a GPU?** No. B1 and B3 run fine on CPU; B2 is easier with a GPU but can be scaled down.
- **Can I pick different model pairs?** Ask first; we want coverage of Lectures 1–6.
- **How much code is expected?** Minimal and correct beats large and brittle.
- **Can I use existing libraries (e.g., diffusers)?** For this HW, prefer implementing core logic yourself.

Reminder: Connect Back to the Course

Your report should explicitly reference lecture concepts when interpreting results:

- Likelihood vs implicit modeling (Lecture 0)
- Scores and denoising (Lecture 1)
- ODE transport and conditional expectation (Lecture 2)
- ELBO terms and posterior collapse (Lecture 3)
- Jacobians and tractable determinants (Lecture 4)
- Minimax dynamics and mode collapse (Lecture 5)
- Factorization and masking (Lecture 6)